Silas Bambil

Statistics and Quantitative Risk

404-468-7274 sdbambil98@gmail.com GitHub: sdbambil1

Education

Georgia State University **Expected Spring 2028**
MS Mathematics, concentration in Statistics *Atlanta, Georgia*

Georgia State University **May 2025**
MS Quantitative Risk Analysis and Management *Atlanta, Georgia*

Kwame Nkrumah University of Science and Technology **July 2020**
BSc Actuarial Science *Kumasi, Ghana*

Experience

Georgia State University **Aug 2026–Present**
Graduate Lab Assistant *Atlanta, Georgia*

- Teach and support College Algebra (MATH 1111) and Support for College Algebra (MATH 0999).

Osceola County School for the Arts **Jan–Jun 2026**
AP Mathematics Teacher *Osceola County, Florida*

- Taught AP Calculus, AP Precalculus, and AP Statistics; students achieved a **70% AP exam pass rate**.

Georgia State University **Aug 2023–Jun 2025**
Graduate Assistant *Atlanta, Georgia*

- Cleaned applicant records and developed predictors for the **Greenberg Sprint** graduate-enrollment model.
- Supported enrollment-model evaluation and provided statistical analysis and data management for faculty risk projects.

OPH CHEM CONSULT **May 2022–Aug 2023**
Finance Associate *Ghana*

- Examined financial misappropriation risks and community financial challenges; contributed research findings to analytical reports and briefs.

Ghana Statistical Services **Sep 2020–Aug 2021**
Research Assistant *Ghana*

- Contributed to population and economic studies through research, statistical analysis, and data management.

Projects

Credit Risk Model Updating | R 2026

- Compared recalibration and refitting on **three Fannie Mae cohorts** comprising approximately **1.55 million eligible loans**, with separate development, adaptation, and temporal test cohorts.
- Evaluated updating and regularization across **22,500 simulated datasets** and five validation allocations, measuring performance on independent test samples.
- Examined how label budgets, shift mechanisms, and regularization changed the relative performance of updating policies.

Fire Loss Modeling and Tail Diagnostics | R 2026

- Analyzed approximately **1.25 million positive-loss records** from NFIRS 2016–2025 using Peaks-Over-Threshold and Generalized Pareto models to estimate Value-at-Risk and Expected Shortfall.
- Compared empirical and fitted expectations within bounded loss layers; assessed uncertainty by incident-cluster bootstrap and examined sensitivity to exceptional records.

shiftadapt | R Package In development

- Implemented **four updating strategies**: retention, intercept recalibration, intercept-and-slope recalibration, and full refitting.
- Separated fitting, validation, and test samples for minimum-log-loss and paired one-standard-error selection.

Publications

- *Selecting and Estimating Credit-Risk Model Updates Under Distribution Shift*. Working manuscript, 2026.
- *Tail Model Diagnostics Using Expected Shortfall and Bounded Loss Layers with an Application to Fire Losses*. Working manuscript, 2026.

Technical Skills

Software: R, Python, Microsoft Excel.
Methods: Regression, Monte Carlo simulation, bootstrap, calibration, model validation, extreme value theory.
Risk analysis: Credit risk, Value-at-Risk, Expected Shortfall, stress testing, scenario analysis.
FINRA examinations: SIE, January 2025; Series 63, August 2025.